AlphaLink CAM Training Manual

Oct 13, 2026 · @Eduard

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About this manual

This manual teaches AlphaLink CAM by doing: five exercises take one part from an empty screen to a program running on the Harrison Alpha 400T. Work through them in order the first time; each one takes 10 to 20 minutes.

AlphaLink CAM replaces the DOS program ALPHA-T. The programs it writes are byte-for-byte the same as ALPHA-T's, so the lathe's macros treat them as ALPHA-T programs.

You need:

- a PC or Surface with AlphaLink CAM installed (Flatpak `io.github._0wn3r.AlphaLinkCAM`, or `python3 -m alphacam_gtk` from the repository);
- for Exercise 5 only: the serial cable to the lathe (DB9 to the lathe's DB25, wired as in the Alphalink-T manual, with its bridges) and a USB serial adapter.

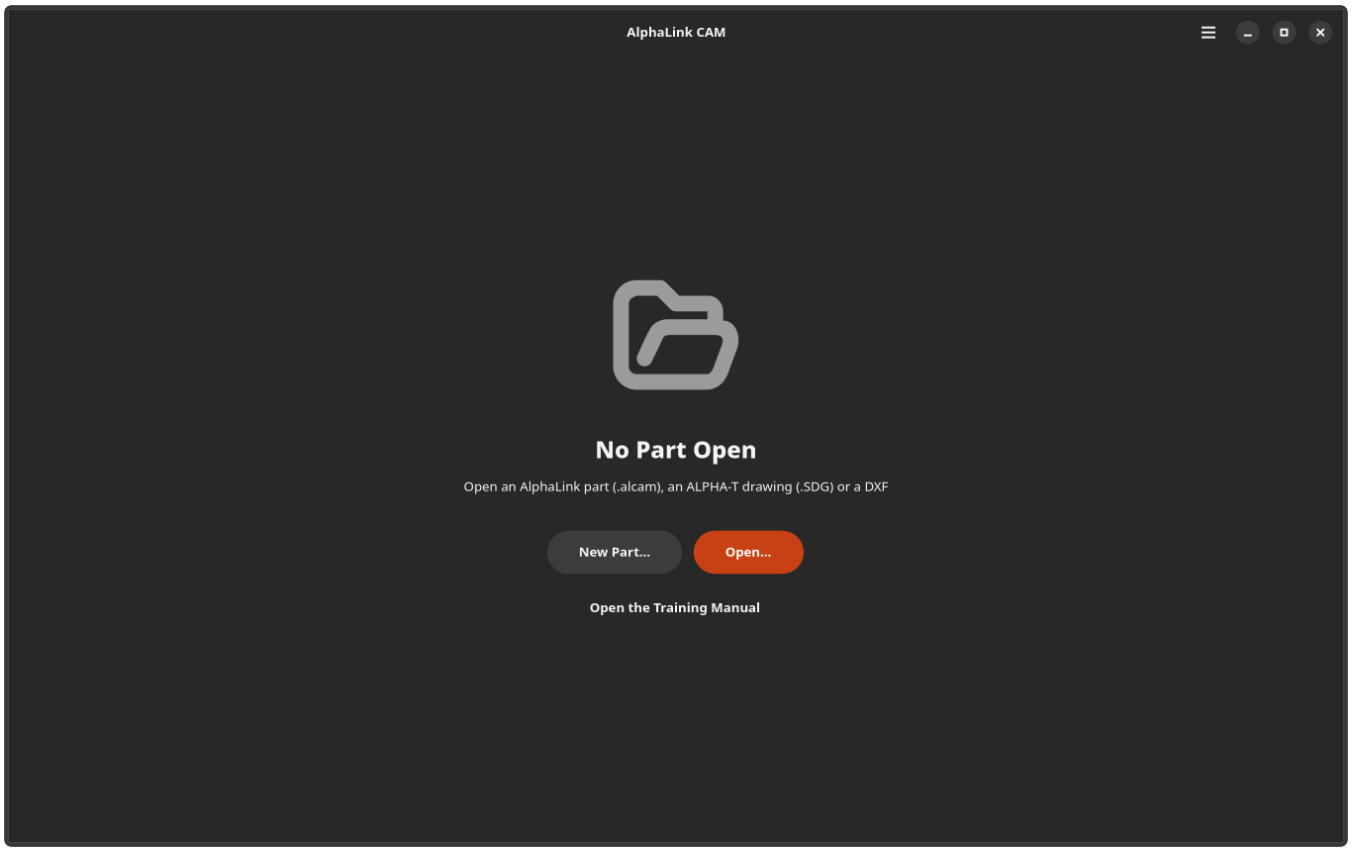
Conventions used in the steps:

- **Tap** means tap with a finger or pen, or click with the mouse.
- Values in `code` are what you type, for example `40,-25` then Enter.
- Button and menu names are shown in **bold** as they appear in English. In Ukrainian the same buttons are in the same places.

- *Note:* lines explain something that is easy to get wrong.

Getting started

Start **AlphaLink CAM** from the applications menu. With no part open you see **New Part…** and **Open…**, and under them **Open the Training Manual**, which opens this manual.



The screen

Area	Where	What it is for
Operations list	left	the machining steps in order; **+** adds one
Page switcher	top	**Part**, **Program**, **Sheet**, **Tools**, **Machine**
Drawing	middle of the Part page	the billet, chuck, part and toolpaths
Simulation buttons	under the drawing	step through the toolpath pass by pass
Tool bar and entry	bottom	Select, Line, Arc, Fillet, Chamfer, Corner, and the `X,Z` entry

Area	Where	What it is for
Operation editor	right, when an operation is selected	that operation's settings
Export PRG	top right	writes the program and its .SD1 listing

On a narrow window (below 1000 px: half a Surface, or a Surface held upright) the operations list becomes its own page, reached with the back arrow, and the page switcher moves to the bottom.

Coordinates

- **X is the diameter** of the part, **Z is the length**. Z0 is the front face; the part runs towards the chuck in -Z.
- The arrows in the bottom-left corner show the directions: +Z right, +X up. The small cross marks X0 Z0.
- Enter real values. The lathe's macros expect Z, feed and pitch doubled in the program, and the app does that for you.

Moving the view

- Drag with one finger, or the mouse, to move the drawing. Pinch with two fingers, or Ctrl+wheel, to zoom.
- The mouse wheel moves up and down, Shift+wheel sideways.
- Double tap, or Ctrl+0, fits the whole part; double tap again goes back.

Note: everything you change can be undone with **Ctrl+Z** or the undo arrow, so it is safe to experiment.

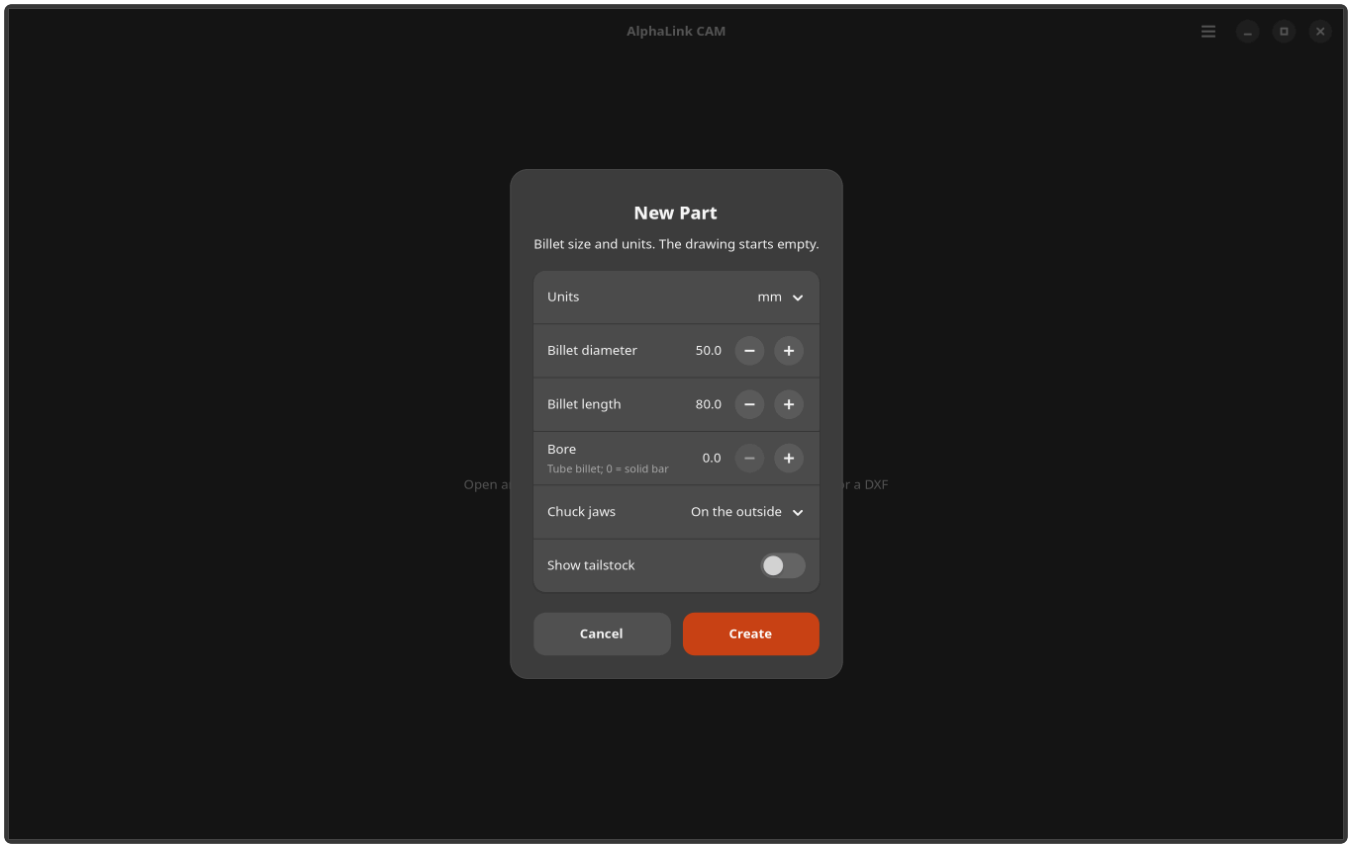
Exercise 1: draw a stepped shaft

You draw a shaft of 40 mm diameter for 25 mm, then 60 mm diameter for 25 mm, from a 100 mm bar, with a chamfer and a fillet. Only the upper half is drawn; the faint lower half is the mirror image.

New part

1. Tap **New Part…** (or Ctrl+N).
2. Set **Units** to mm, **Billet diameter** `100`, **Billet length** `60`, and tap **Create**.

3. The chuck, the dashed billet and the axis line appear. The Line tool is already selected.



Note: New Part also asks for a **Bore** (for a tube billet), which way the **Chuck jaws** grip, and whether to **Show tailstock**. Leave them as they are for these exercises. **Part Settings** in the main menu changes them later, together with the program number shown on the control.

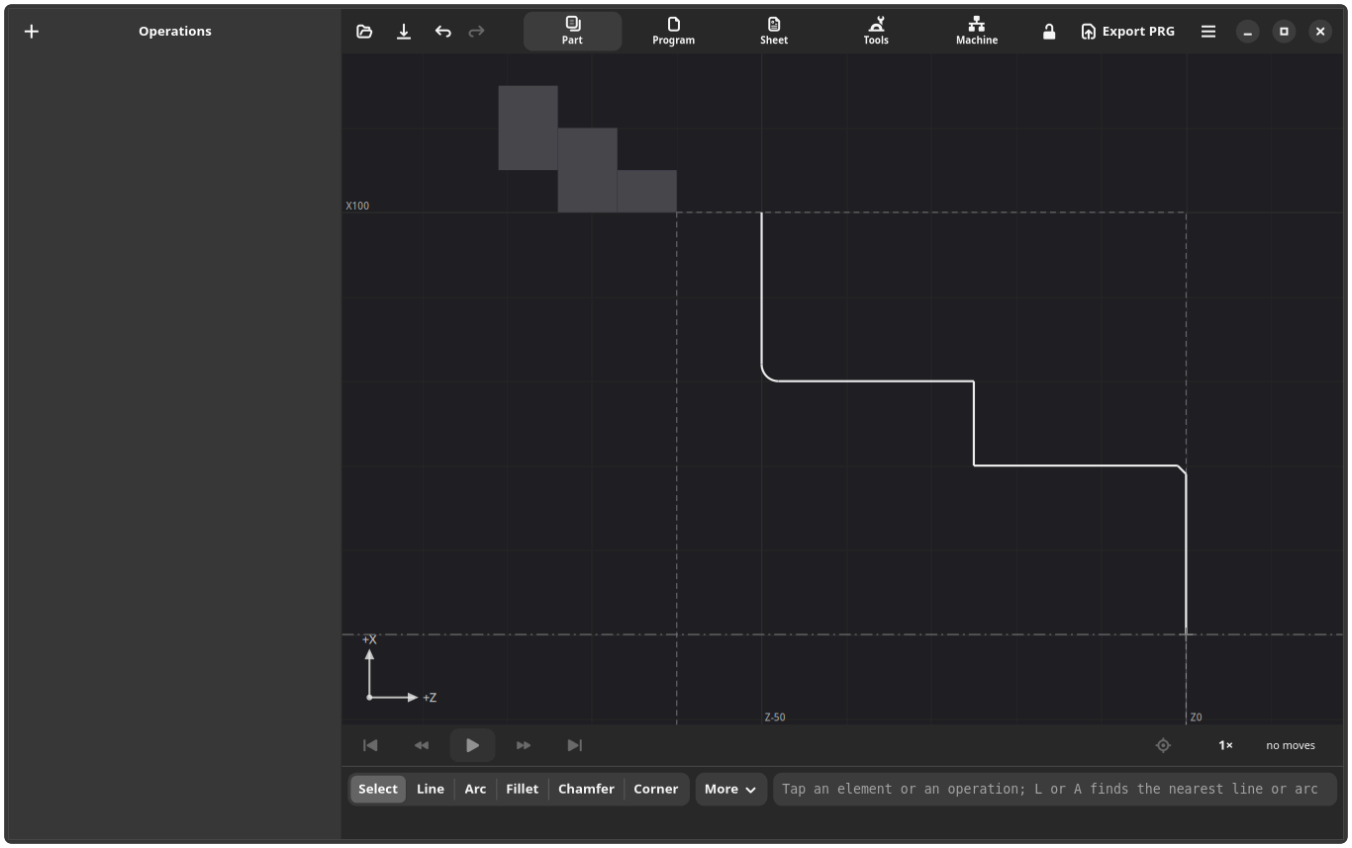
Lines by coordinates

Each point is typed in the entry under the drawing as `X,Z`, then Enter. `@dX,dZ` is relative to the last point.

1. Type `0,0` Enter. This is the start point, on the axis at the front face.
2. Type `40,0` Enter. A line goes up the front face to diameter 40.
3. Type `@0,-25` Enter. The 40 mm diameter, 25 mm long.
4. Type `60,-25` Enter. The shoulder up to diameter 60.
5. Type `60,-50` Enter. The 60 mm diameter.
6. Type `100,-50` Enter. The back shoulder out to the bar diameter.
7. Tap **Done** (or press Esc) to finish the line.

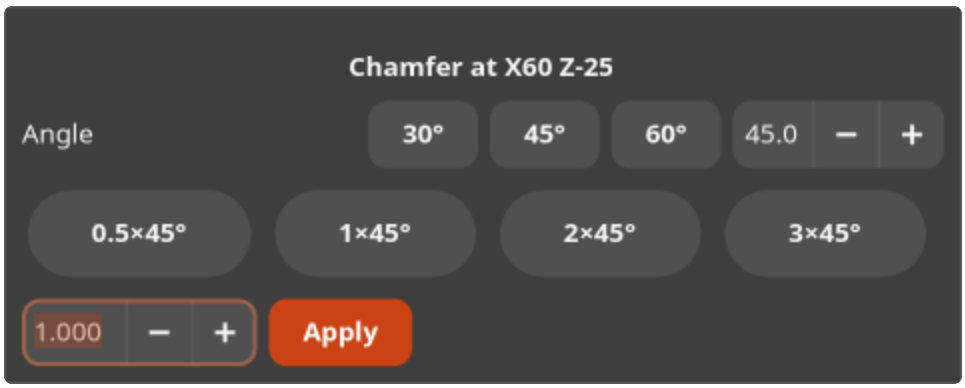
Note: Z is negative towards the chuck. Typing `40,25` instead of `40,-25` draws in front of the part.

Note: you can also tap points on the drawing. Taps snap to line ends, the axis and a 1 mm grid, and lock to Z, X or 45 degrees, so typed values are only needed where the grid does not reach. With a finger, a chip on the last line shows where it ends; tap the chip to correct that point on the keypad. Tapping the first point again closes the profile and finishes the line.



Chamfer and fillet

1. Tap **Chamfer**, then tap the corner at X40 Z0. In the popup keep the angle at 45° and tap **1×45°**.
2. Tap **Fillet**, then tap the inside corner at X60 Z-50. Tap **R2**.
3. Tap **Select**. Tap any line to see its ends; Delete erases it, and Ctrl+Z brings it back. Drag an end handle to move that point; the lines joined to it follow.



Note: a chamfer's size is measured along the line that runs along Z, and its angle is from that line. 2×30° on a shoulder is 2 mm along the diameter and 1.155 mm down the face.

Note: to change a line you already drew, select it and tap the pencil next to the bin. Type the new end points; the lines it meets move with it, so the profile stays joined.

Note: a long press on an element, or a right click, opens a menu with **Edit Values…**, **Delete**, **Zoom Here** and **Fit Part**. From the keyboard, the Menu key or Shift+F10 opens it at the cursor.

Save

Press Ctrl+S and save the part as `shaft.alcam`. You use it in the next exercises.

If you used ALPHA-T: its one-letter keys work here too. **X** draws by X,Z, **F** puts a radius on the nearest corner, **K** trims a corner, **Z** zooms and **R** redraws. The full list is in the Reference section.

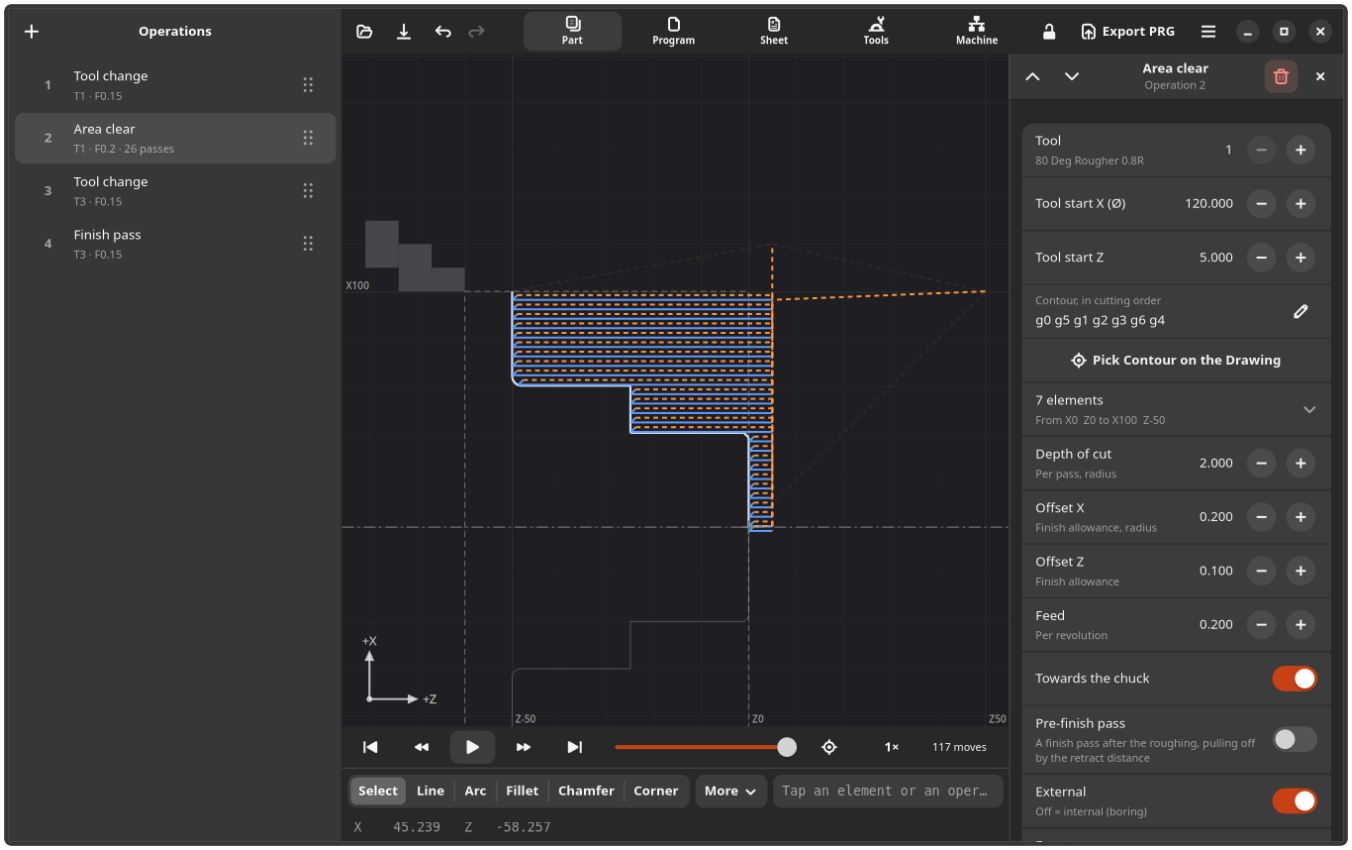
More at the end of the drawing bar has ALPHA-T's construction commands: circle, tangents, tangent at an angle, parallel line, split arc, blend arc and mirror. Under them are **Standard Thread…** (Exercise 3), and **Chamfer Outside Corners…** and **Round Outside Corners…**, which put one size on every outside corner at once; change single corners afterwards with the Chamfer and Fillet tools. To stop any drawing command and choose another, press **Esc** or tap another tool.

Exercise 2: rough and finish

You rough the shaft with the 80° rougher (tool 1) and finish it with the 55° finisher (tool 3). Open `shaft.alcam` from Exercise 1.

Roughing

1. Tap **+** above the operations list and choose **Tool Change**. Set **Tool** to `1`. The name under it (80 Deg Rougher 0.8R) comes from the Tools page.
2. Tap **+** and choose **Area Clear**. The new operation already uses tool 1 and the outer profile as its contour.
3. In the editor set **Depth of cut** `2` (per pass, on radius) and **Feed** `0.2`.
4. Check **Tool start**: X `120`, Z `5` keeps the tool clear of the bar; set it if it shows the billet diameter.
5. The passes appear on the drawing: blue lines are cuts, orange dashed lines are rapid moves.

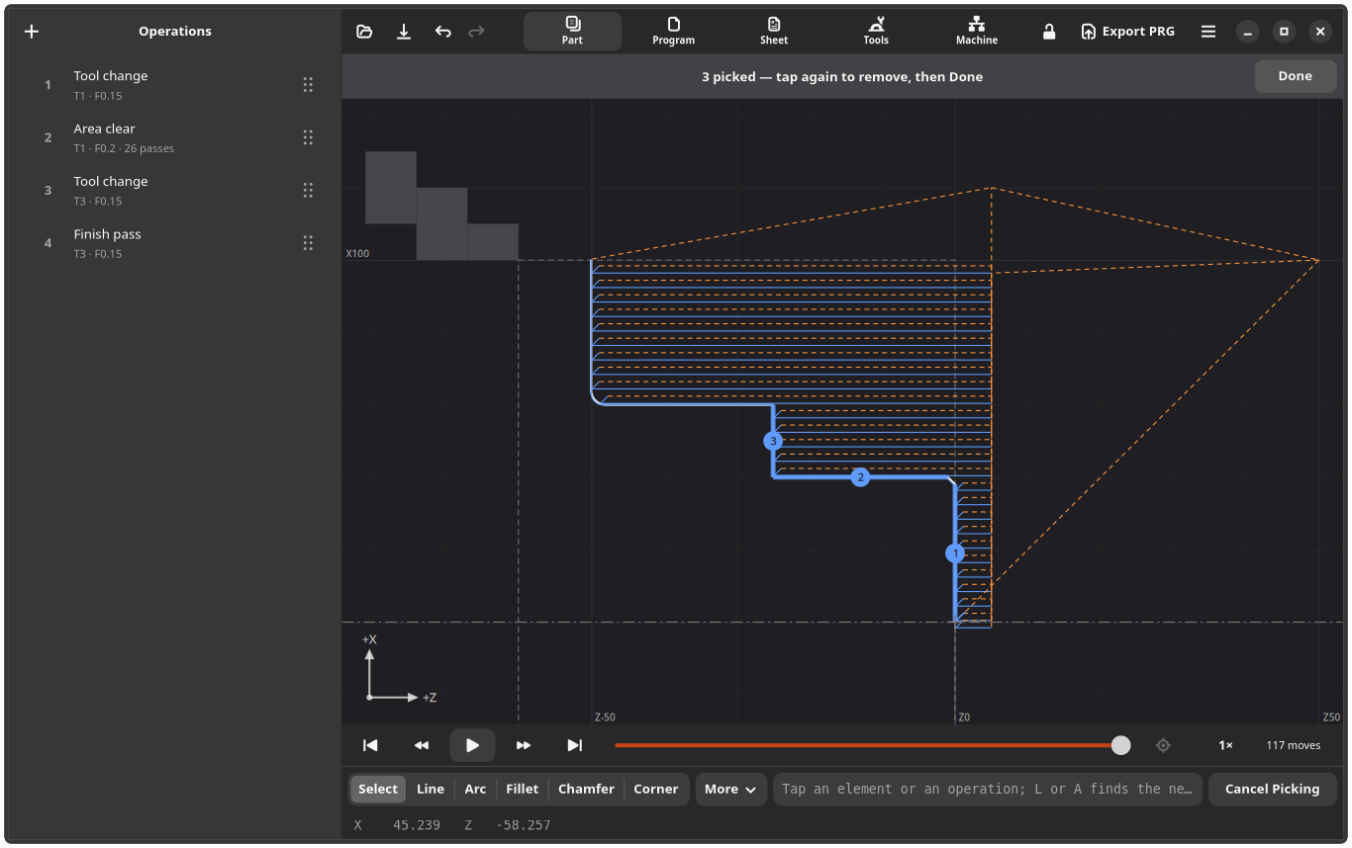


Note: **Tool** is the tool memory slot on the control. The turret is manual, so at a tool change the lathe stops at the tool change point (X100 Z50 unless you change it) for you to fit that tool.

Picking the contour yourself

The contour is the drawing elements the tool follows, in cutting order. New operations take the outer profile, leaving out thread lines and construction lines; to cut only part of it:

1. In the editor tap **Pick Contour on the Drawing**.
2. Tap the elements in the order the tool should follow them. Numbered badges show the order; tap one again to remove it.
3. Tap **Done** in the banner, or press Enter.



Finishing

1. Tap **+**, choose **Tool Change**, set **Tool** `3`.
2. Tap **+**, choose **Finish Pass**. Leave **Offset X** and **Offset Z** at 0.
3. To leave stock for the finish pass, go back to the Area Clear and set **Offset X** `0.2` and **Offset Z** `0.1`.

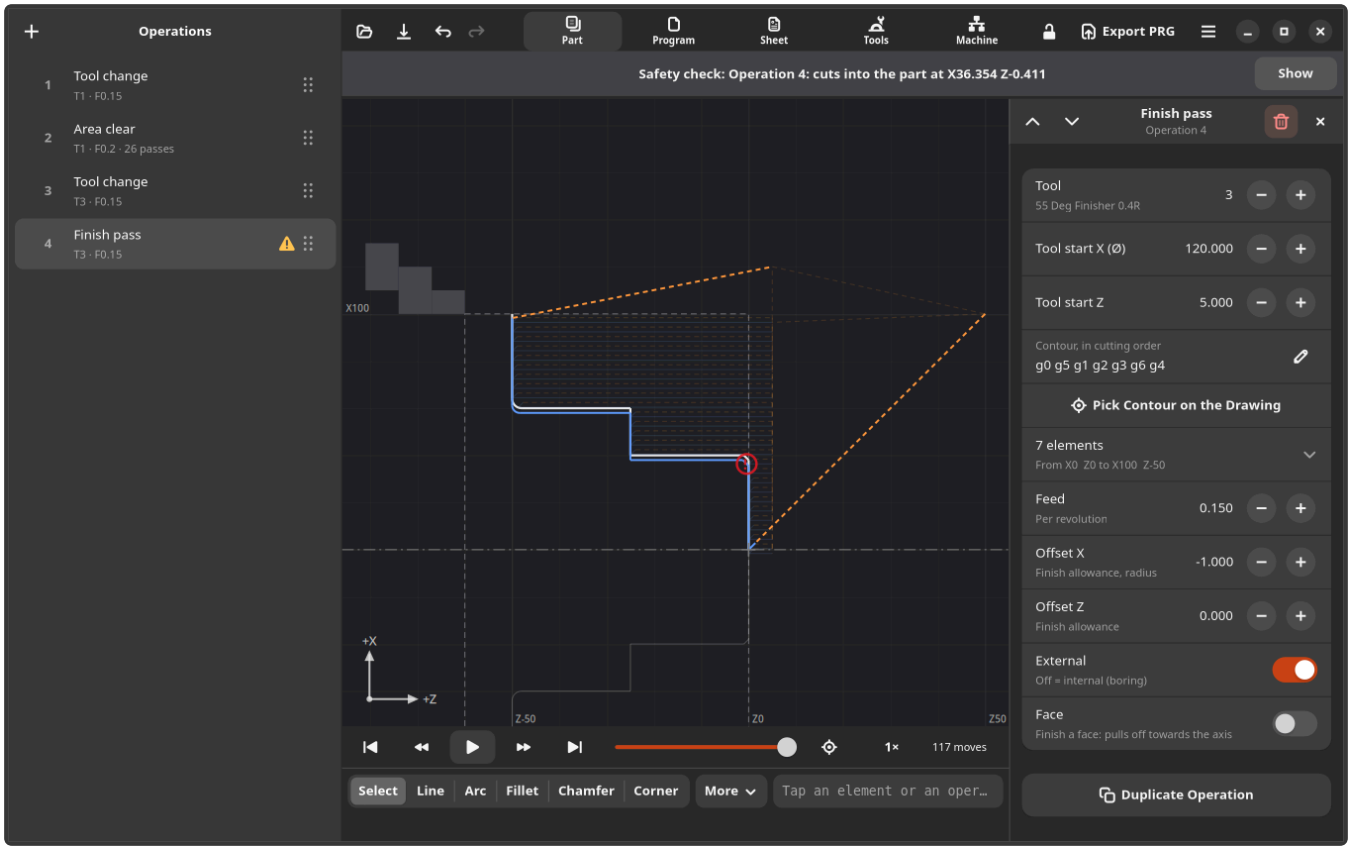
Simulate

Use the buttons under the drawing: ⏮ first move, ⏪ previous pass, ▶ play or pause, ⏩ next pass, ⏭ show everything. The label shows the move number, the operation, the pass and the program line. Tap a toolpath to select its operation.

The **1×** button changes the playback speed, and the button next to it jumps to the start of the selected operation. Tapping a toolpath jumps the simulation there too. To change the order of the operations, drag a row by its handle.

Note: if an operation cannot be made into a program, a banner at the top says which one and why, and **Export PRG** is greyed out until it is fixed.

The app also checks every move before you export. If a move cuts into the part, goes into the chuck jaws or the tailstock centre, or dives steeper than the tool's clearance allows, a banner names the operation and the place. **Show** marks the place with a red ring, and the operation row gets a warning sign. These are warnings only: **Export PRG** still works, so a deliberate program is never blocked.



Save the part (Ctrl+S).

Shortcut: **+ ▸ Templates ▸ Shaft: rough T1, finish T3** adds this whole exercise's four operations at once, with the contour picked from your drawing. **Save Operations as Template…** in the same menu keeps your own sequence for the next part.

Exercise 3: grooving and threading

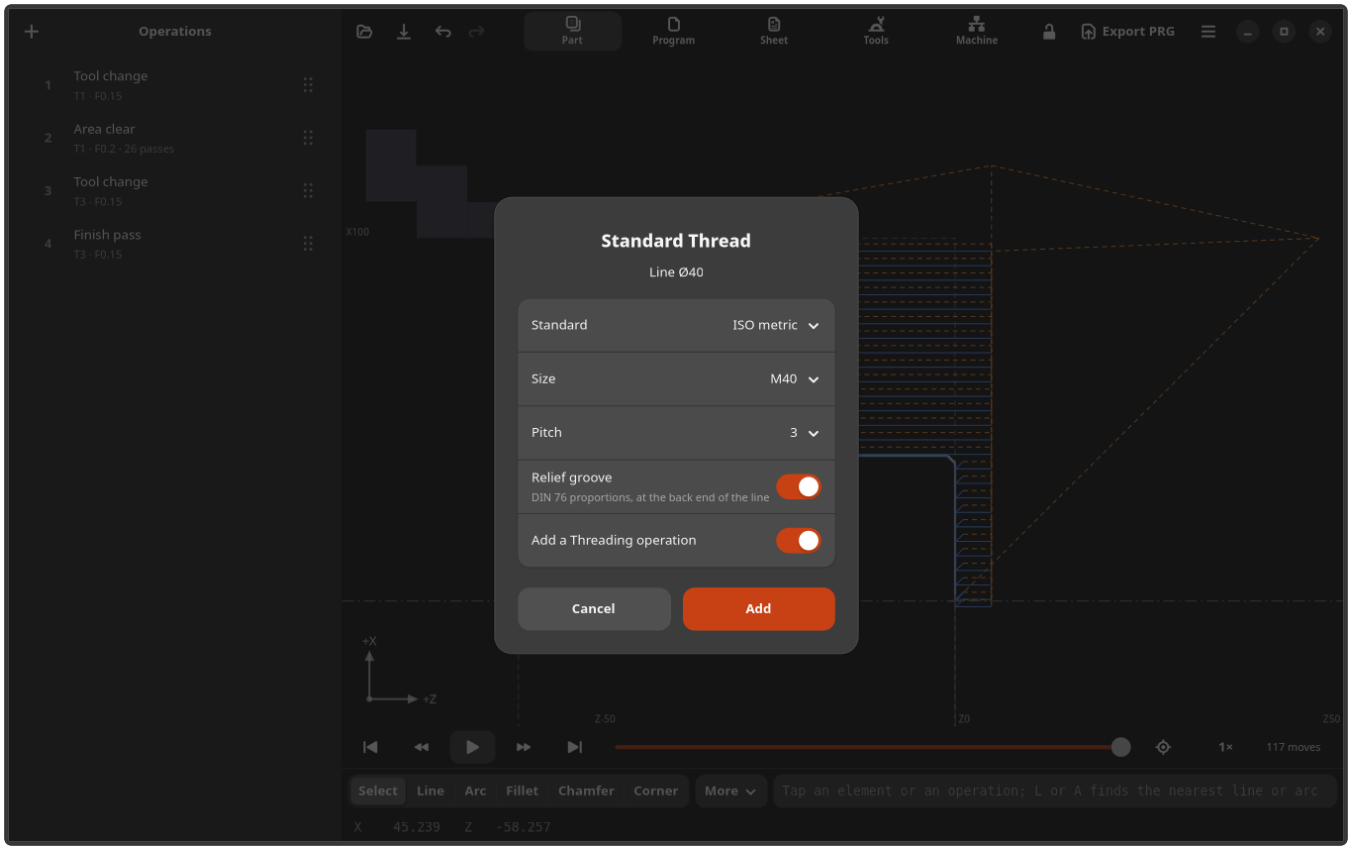
You cut a groove and an M40×1.5 thread. Grooving follows the drawn profile, so this exercise starts a new, simpler part.

Draw the part

1. **New Part…**: billet diameter `60`, length `60`.
2. Line tool, then type each point and Enter: `0,0`, `40,0`, `40,-20`, `34,-20`, `34,-26`, `40,-26`, `40,-40`, `60,-40`. Tap **Done**.
3. The thread needs its root diameter as a line too. Type `38.2,0` Enter, `38.2,-18` Enter, then **Done**. This line is not cut; it only tells the thread how deep to go.

Shortcut: instead of typing the 38.2 line, tap **Select**, tap the 40 mm line from Z0 and choose **More ▸ Standard Thread…**. Choose M40 and pitch 1.5, and switch **Relief groove** off, because this part has its own groove. The app draws the minor-diameter line (Ø38.16) and adds the Threading operation with its tool change. On a plain diameter, leave the relief groove on: it is cut at the back end of the line.

Note: under **Standard** the dialog also has UNC and UNF inch threads, 1/4″ to 2″. On a mm part they are converted to mm, and an inch part starts on UNC.



Grooving

1. **+ ▸ Tool Change**, **Tool** `16` (2 mm groove/parting tool).

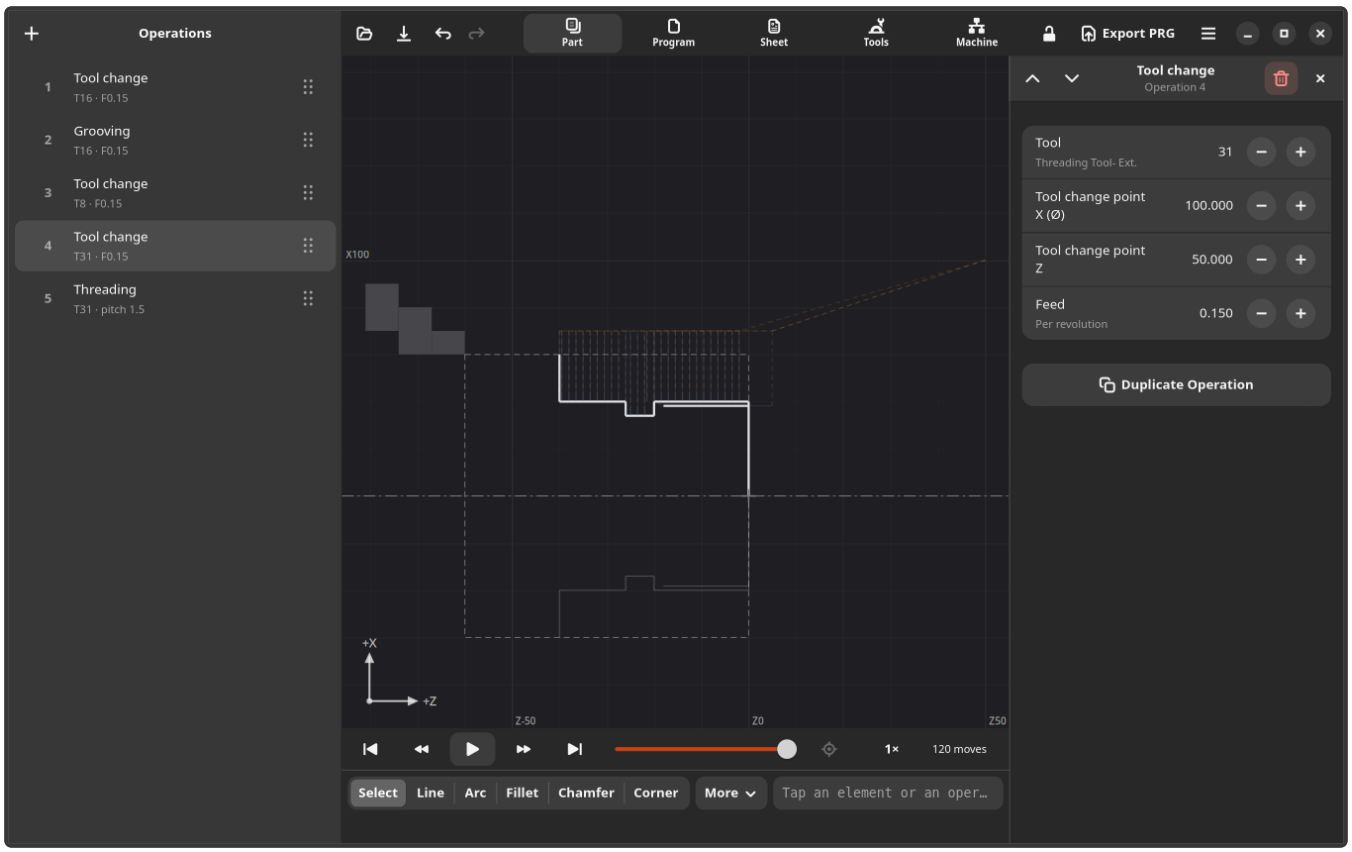
2. **+ ▸ Grooving**. The app picks the contour for you: the 40 diameter before the groove, the three groove sides, and the 40 diameter after it. Check the highlighted elements, or pick them yourself.
3. Set **Tip width** `2`, **Stepover** `1.5`, **Peck depth** `0` (plunge in one go). Leave **Mode** on area and **Datum** on left.
4. Optional: **Dwell** `1` waits one spindle turn at the bottom of each plunge for a clean floor.

Note: **Datum** says which corner of the tip the tool memory measures: left is the -Z corner, as on ALPHA-T.

Note: when the tool in the turret does not suit a new Grooving, Threading or Drilling operation, the app adds a Tool Change to a suitable tool before it. Check that its number is the right slot on the Tools page.

Threading

1. **+ ▸ Tool Change**, the threading tool's slot.
2. **+ ▸ Threading**. The app sets **Outer diameter line** to the 40 mm line from Z0 and **Inner diameter line** to the 38.2 line; check them.
3. **Pitch** `1.5`. Either **First cut depth** `0.2` (cut n goes to 0.2 × √n), or **Number of cuts** `8`.
4. **Lead in** stays at 5 mm: every cut starts 5 mm in front of the thread to get up to speed. **Lead out** `0`.
5. Set **Spring passes** to `1` for one more cut at the final diameter.



Note: keep **Towards the chuck** on. ALPHA-T's AWAY threading starts every cut inside the thread, so the app refuses it and says why.

Exercise 4: face, drill and bore

You face a part, drill a 20 mm hole and bore it out to a stepped bore.

Face clear

Open `shaft.alcam` again.

1. **+ ▸ Face Clear**. It cuts across the faces towards the axis, one depth of cut at a time along Z.
2. **Depth of cut** `2`, feed as you like. The passes stop on each diameter of the drawing.
3. Optional: **Pre-finish pass** adds a finishing pass along the contour after the roughing, in the same operation.

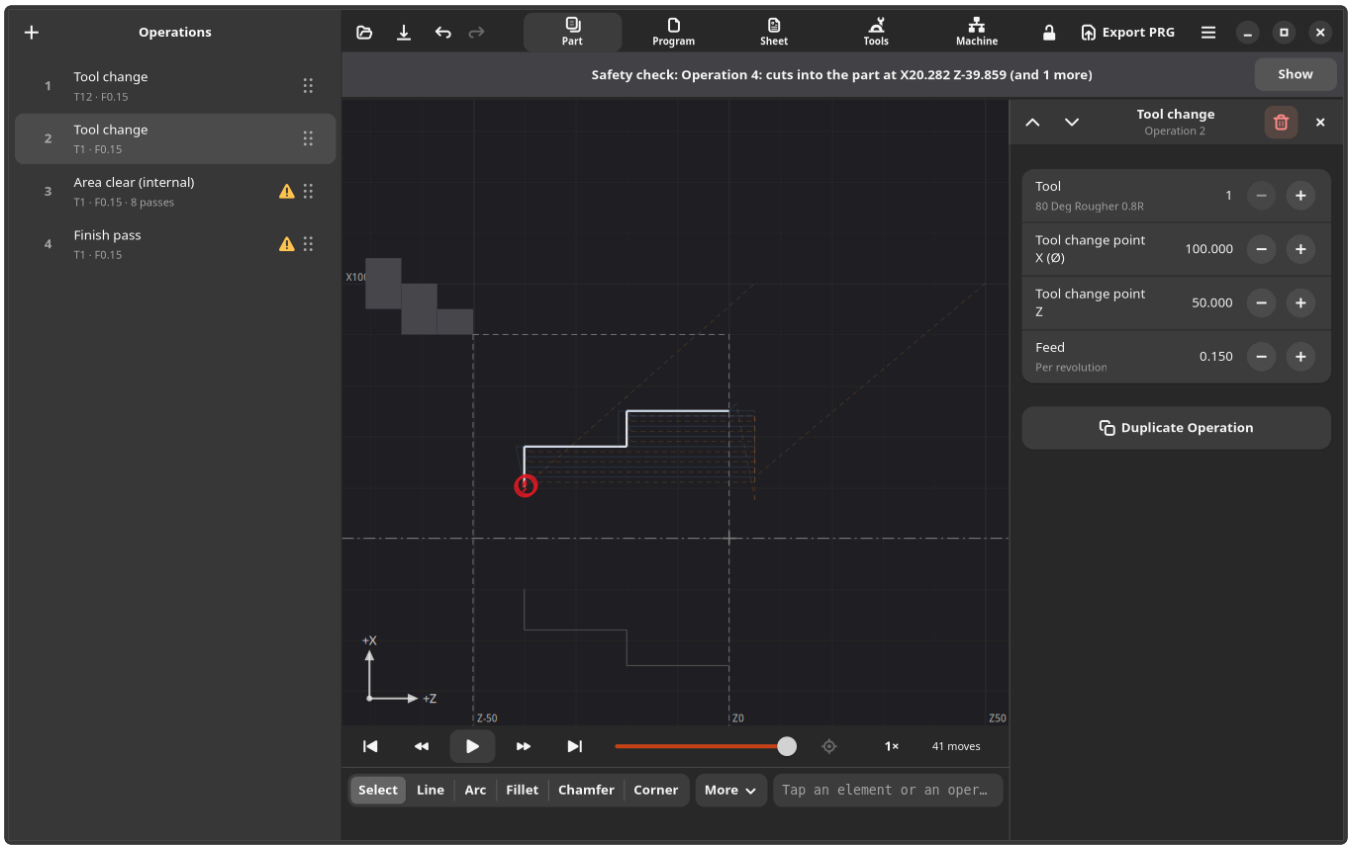
Note: on Area Clear, switching **Front** off gives the same face-wise passes with the area clear retract.

Drilling

1. **New Part…**: billet diameter `80`, length `50`.
2. **+ ▸ Tool Change**, the drill's slot. **+ ▸ Drilling**.
3. **Tool start** X `0`, Z `5`. **Hole bottom Z** `-42` (an absolute Z, not a depth). **Peck** `10` drills 10 mm at a time and backs off 2 mm between pecks; `0` drills in one go. **Dwell** waits at the bottom.

Boring

1. Draw the bore with the Line tool: `50,0`, `50,-20`, `36,-20`, `36,-40`, `20,-40`, then **Done**. The 20 mm hole from the drill is the start.
2. **+ ▸ Tool Change**, **Tool** `12` (55° boring bar 0.4R).
3. **+ ▸ Area Clear**. Turn **External** off: the editor title shows *(internal)*. Set **Tool start** X `15`, Z `5`, inside the drilled hole.
4. **+ ▸ Finish Pass**, **External** off as well.
5. Simulate. The passes now feed outwards from the axis and pull off towards it.



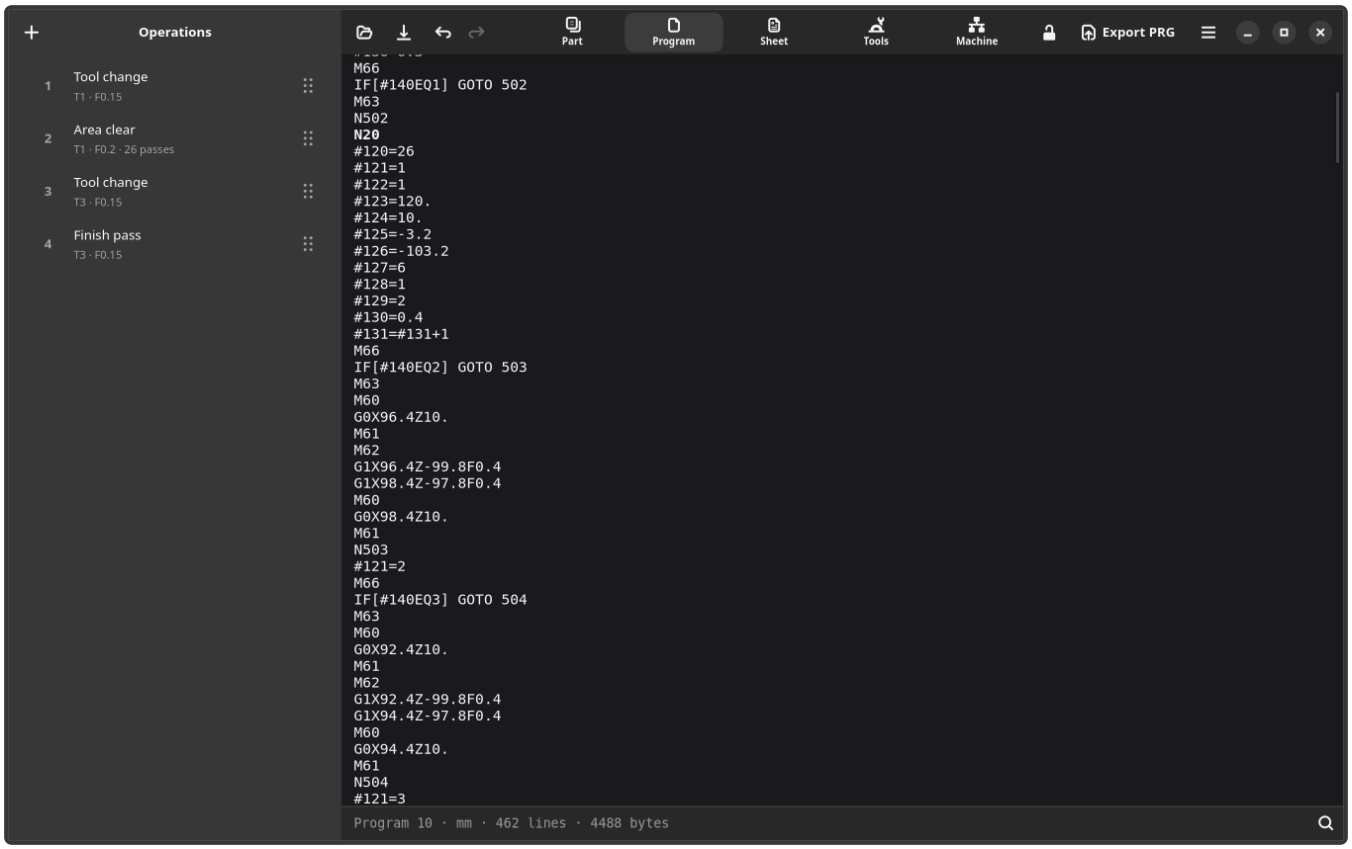
Note: the tool start of an internal operation must be inside the hole. If X is larger than the bore, the first rapid move goes through the part.

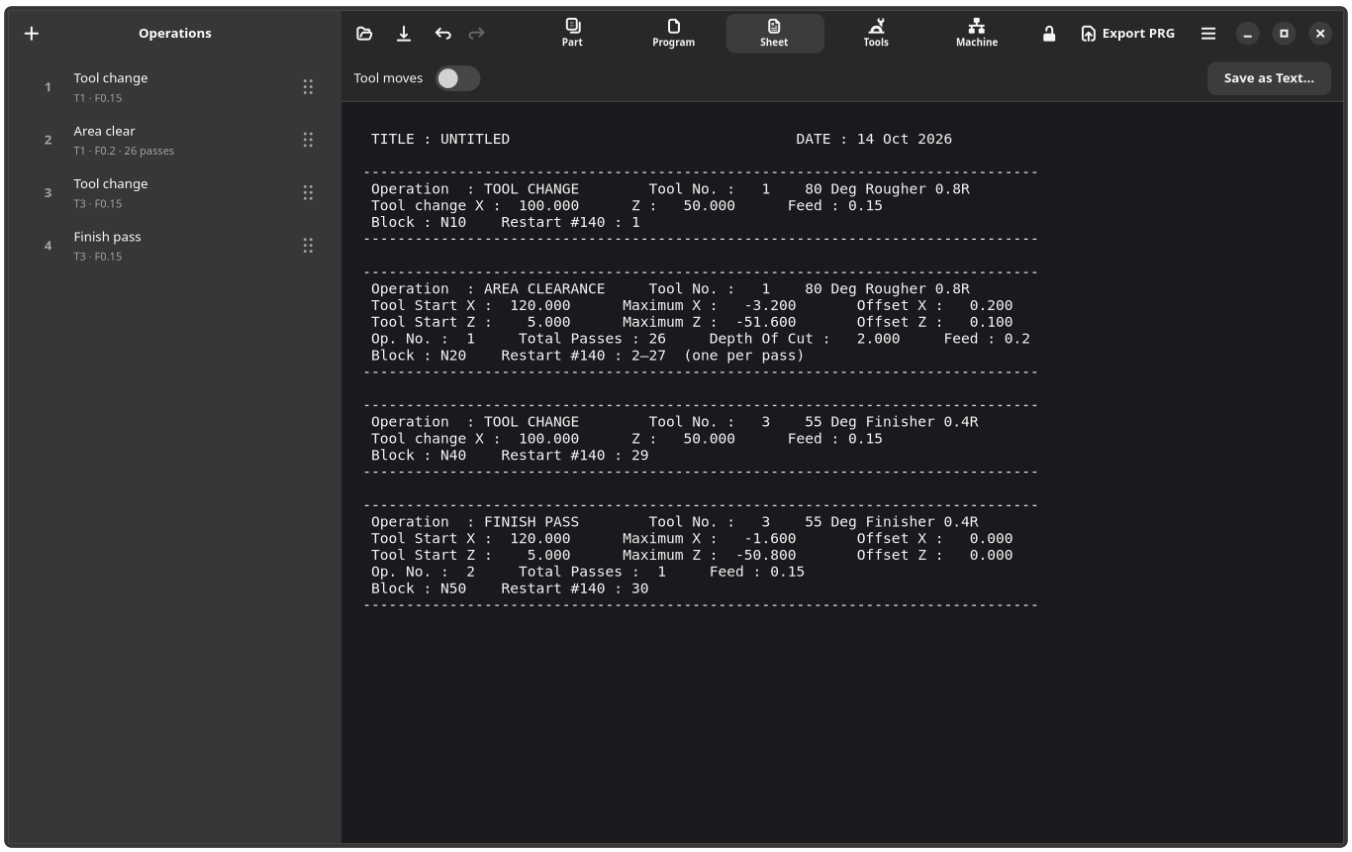
Exercise 5: check, export and send

You check the shaft program from Exercise 2, save it as files, and load it into the lathe. Open `shaft.alcam`.

Check

1. **Program** page: the program exactly as the lathe gets it. Tap an operation in the list to highlight its lines; pinch to change the text size. Ctrl+F, or the search button, finds text in it; type N10 to go to block 10.
2. **Sheet** page: the setup sheet, with each operation's tool, start point, limits, passes and feed, like ALPHA-T's print, with each block's restart number (#140) and any tool warnings. Turn on **Tool moves** to list every move. **Save as Text…** writes it to a file for the shop floor. The bar shows the cycle time estimate, and Save as PDF… writes a printable sheet with the part, the tools in loading order and your notes for the operator.
3. **Tools** page: check that each tool number used is the slot that holds that tool's offsets on the control.



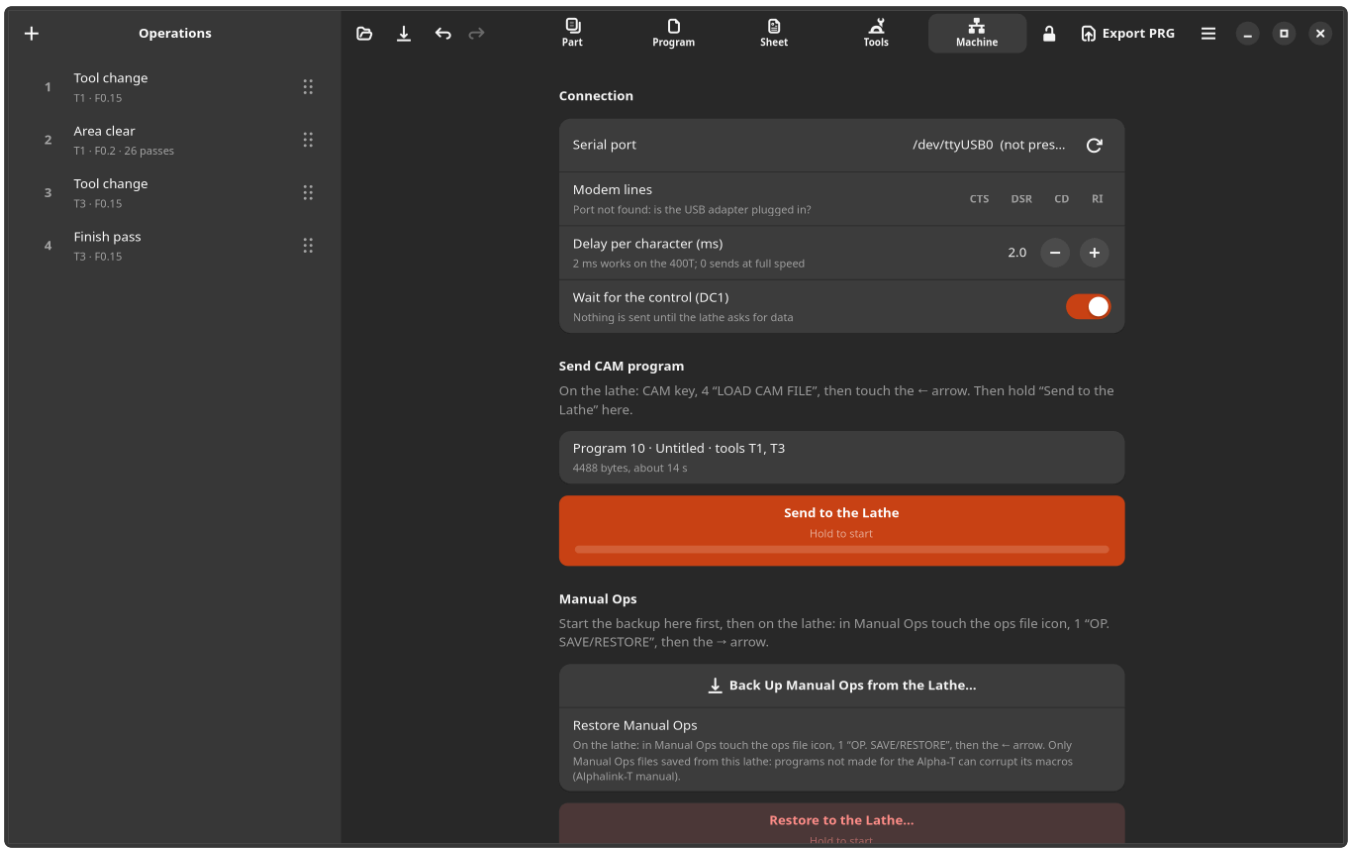


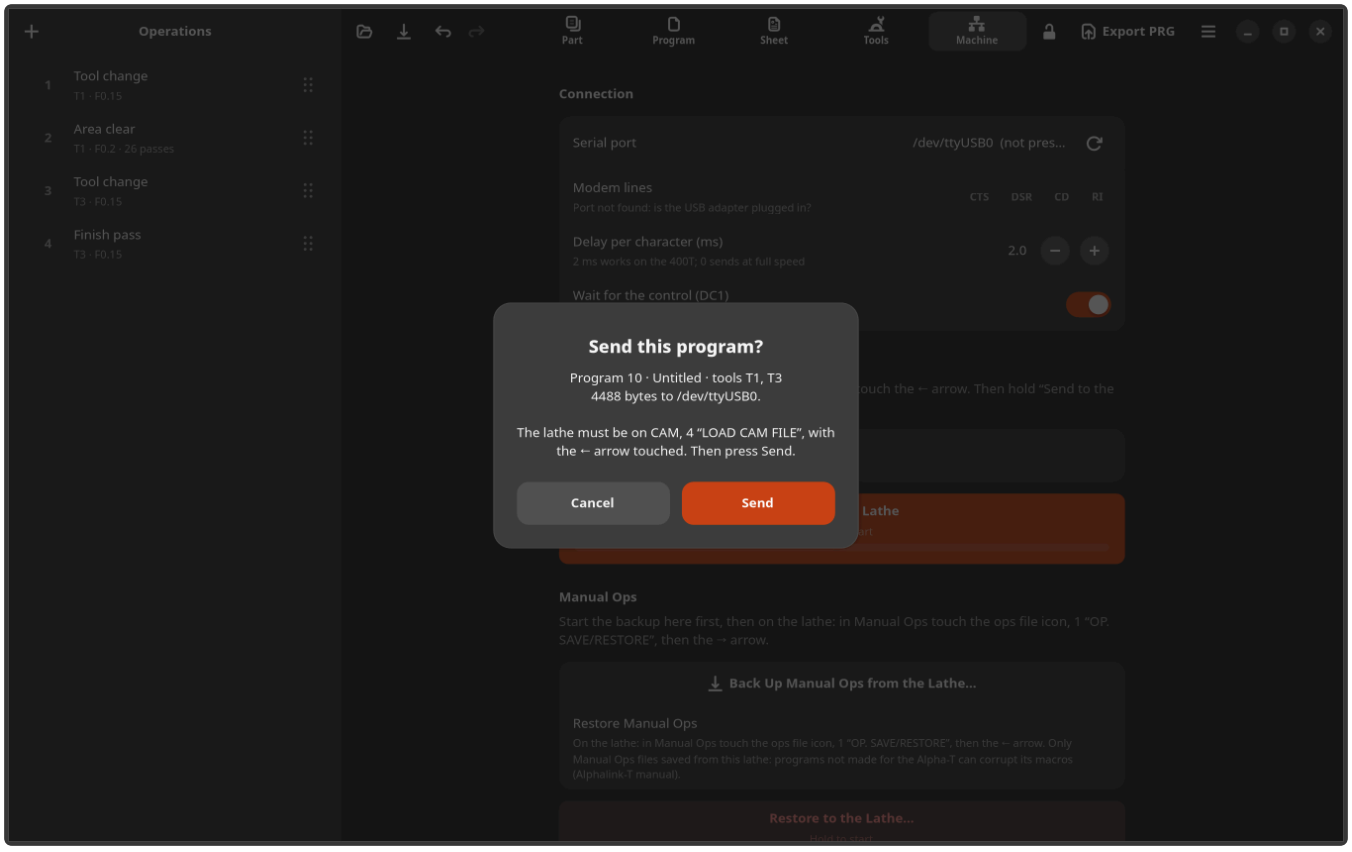
Export

Tap **Export PRG** (Ctrl+E) and save as `SHAFT.PRG`. The app writes `SHAFT.PRG` and, next to it, `SHAFT.SD1`, the same pair ALPHA-T writes. Next time, export opens the same folder and suggests the part's name, cut to ALPHA-T's 8 characters.

Send to the lathe

1. Connect the serial cable and the USB adapter. On the **Machine** page choose the **Serial port** (usually /dev/ttyUSB0). Keep **Delay per character** at 2 ms and **Wait for the control (DC1)** on.
2. On the lathe: press the CAM key, choose **4 LOAD CAM FILE**, and touch the ← arrow.
3. In the app, hold **Send to the Lathe** until it fills, check the summary (program number, tools, size) and tap **Send**.
4. The transfer window shows the bytes sent. When it finishes, the program number shows on the control.





Note: the app never starts the machine. Check the program on the control, and run a new program the first time with the feed override turned down.

Back up Manual Ops

Do this once before you load anything new, and keep the file.

1. In the app tap **Back Up Manual Ops from the Lathe…** first. It waits for data.
2. On the lathe, in Manual Ops, touch the ops file icon, choose **1 OP. SAVE/RESTORE** and touch the → arrow.
3. Save the file when the app asks. To put it back later, use **Restore to the Lathe…** and the ← arrow on the lathe.

Note: restore only files saved from this lathe. Programs not made for the Alpha-T can corrupt its macros.

Cable test

If nothing arrives, unplug the cable from the lathe, bridge pins 2 and 3 of the adapter, and tap **Run Loopback Test**. It proves the adapter and port work; the rest is the cable or the lathe's settings.

Reading the program

The .PRG is not plain G-code: it sets variables that the lathe's Harrison macros (B703-0703) read, then calls those macros with M-codes. Values marked *inferred* come from comparing many ALPHA-T programs; Harrison's macro listing would confirm them.

Note: in the .PRG every Z value, feed and thread pitch is **twice** the real value (Z50 is written `#124=100.`, feed 0.15 as `#130=0.3`); the macros halve them. X is a diameter and is written as it is. The .SD1 listing is not doubled.

Program header

Line	Meaning
`%`	start and end of the file
`O2`	program store on the control; always O2
`#132=10`	program number shown on the control, set in Part Settings (confirmed on the 400T)

Line	Meaning
`#140=0`	restart point: 0 runs from the start; n jumps to restart label N(501+n) (*inferred*: set by SELECT REQUIRED PASS)
`IF[#1006EQ1]GOTO[[#129+1]*10]`	if control input #1006 is on, continue at the next block (*inferred*: run mode switch)
`IF[#1005EQ1]GOTO[#129*10]`	if input #1005 is on, restart the current block (*inferred*)
`#131=#131-1` / `#131=0`	operation counter, reset at the start (*inferred*)
`IF[#1009EQ1]GOTO[#129*10]`	if input #1009 is on, restart the current block (*inferred*)
`M71` / `M70`	metric / inch

Block variables

Each operation is a block `N10`, `N20`, … that sets these, then runs its passes.

Variable	Meaning	Example
`#120`	number of passes in the operation	`#120=26`
`#121`	pass number, set again before each pass	`#121=3`
`#122`	tool number = tool memory slot on the control	`#122=3`
`#123`	tool start X (in a tool change block: tool change point X)	`#123=120.`
`#124`	tool start Z ×2 (tool change point Z ×2)	`#124=10.` = Z5
`#125`	limit X: lowest drawn diameter minus 4× nose radius	`#125=-1.6`
`#126`	limit Z ×2: lowest drawn Z minus 2× nose radius	`#126=-101.6` = Z-50.8
`#127`	block kind (table below)	`#127=6`
`#128`	always 1 (*meaning unknown*)	`#128=1`
`#129`	block number: 1 for N10, 2 for N20, …	`#129=2`

Variable	Meaning	Example
`#130`	feed ×2; for threading the pitch ×2	`#130=0.3` = F0.15
`#131`	`#131=#131+1` counts operations (*inferred*)	

Block kinds (#127)

#127	Block
1	tool change
6 / 7	area clear / internal area clear (also area clear FACE on a bore)
8	face clear (also external area clear FACE)
9	grooving
10 / 11 / 12	pre-finish after area clear / internal area clear / face clear
14 / 15	finish pass / internal finish pass
17	grooving finish mode
18	drilling
20 / 21	threading / thread spring pass

Codes in the moves

Code	Meaning
`M60` … `M61`	around every rapid move `G0 X Z` (*inferred*: the macros check the move)
`M62`	start of a group of feed moves
`G1 X Z F`	feed move; `G2` / `G3` arc clockwise / anticlockwise with `I` (X to the centre, as a diameter) and `K` (Z to the centre, ×2)
`G32 X Z F`	thread cut; `U` `W` the pull-out, as increments

Code	Meaning
`G4 P`	dwell, P = revolutions × 1000
`M63`	after a restart label, before the first move of a pass (*inferred*)
`M64`	end of an operation's passes
`M65`	end of the program
`M66`	start of a block or pass (*inferred*: updates the display)
`N5nn` + `IF[#140EQn] GOTO 5nn`	restart labels, one per pass and block

The X and Z values in the moves are the **imaginary tool tip** path: the app has already allowed for the tool nose radius, so straight lines along the axes sit exactly on the drawing and arcs and slopes are shifted.

The .SD1 listing

The .SD1 that comes with the .PRG is ALPHA-T's own record of the operations, in real units: `G5` tool start, `G115` limits, `G70` finish, `G71` area/face clear (E100 external, E-100 internal, E401 face, E101 FACE direction, H = passes), `G75` grooving, `G76` threading (Q = thread depth, D = first cut), `G74` drilling, `G99` tool change, then the nose-centre path. ALPHA-T reads its operations back from this file, and so does AlphaLink CAM when you open an .SDG drawing.

Reference

Operations

Operation	What it cuts	Main settings
Area Clear	roughing passes along Z	depth of cut, offset X/Z, external/internal, Front (off = face-wise passes), pre-finish pass, follow contour, retract distance
Finish Pass	one pass along the contour	offset X/Z, external/internal

Operation	What it cuts	Main settings
Face Clear	roughing passes across the faces	depth of cut, offset X/Z, pre-finish pass
Grooving	plunges along a drawn groove	tip width, stepover, peck depth, full retract, dwell, datum, finish offset X/Z, Z shift and repeats, mode area/finish, external/internal
Threading	thread cuts between two diameter lines	pitch, first cut depth or number of cuts, lead in/out, spring passes, infeed angle, external/internal
Drilling	hole on the axis	hole bottom Z, peck, dwell, retract distance, full retract
Tool Change	stops at the tool change point for the next tool	tool slot, tool change point, feed
Dwell	waits a number of spindle turns	revolutions
Move / Rapid	a move to a point	X, Z, rapid or feed

Finish Pass also has **Face** (finishing a face, pulling off towards the axis), and Grooving with **Front** off cuts a groove drawn into a face (face grooving). The app refuses ALPHA-T's AWAY (area, face, thread), because ALPHA-T's own programs for it cut past the shoulder, into the diameter or start inside the thread. A contour picked from the chuck end is cut as if picked from the front.

Keyboard

Keys	Action
Ctrl+N / Ctrl+O / Ctrl+S / Ctrl+Shift+S	new, open, save, save as
Ctrl+E	export PRG and SD1
Ctrl+Z / Ctrl+Shift+Z	undo / redo
Ctrl+0, Ctrl++, Ctrl+-	fit, zoom in, zoom out
Esc	finish the line or command, cancel picking, deselect; the keyboard goes back to the drawing

Keys	Action
Delete	erase the selected element
F1	help
Ctrl+F	search the program; N10 goes to block 10
Shift+F10, Menu key, right click	the drawing's menu: Edit Values, Delete, Zoom Here, Fit Part
Ctrl+?	all keyboard shortcuts

ALPHA-T one-letter keys (Part page)

Key	Action	Key	Action
X	draw line by X,Z	D	draw line by length,angle
+	line by taps	S	start a line at the cursor
G / H / Y	cursor to X,Z / by dX,dZ / by length,angle	P / I	nearest point / intersection
L / A	find nearest line / arc (then - erases)	F / O	radius at a corner / blend arc between two elements
K	trim corner	M / N	measure X,Z / length,angle
T / U	text / erase text	Q	erase the drawing (asks first)
Z / R / V	zoom / redraw / previous view	C	operations menu

Arrow keys move the cursor (PgUp/PgDn change the step), Home centres it, End puts it on X0 Z0, Enter takes it as the next point. While picking a contour, 3 or Enter finishes.

Troubleshooting

You see	Do this
Export PRG is greyed out	read the banner at the top: it names the operation and the reason
Opening an .SDG shows no operations	the .SD1 with the same name must be next to it; use **Choose .SD1…** in the message
Nothing arrives at the lathe	check the port, the CAM screen is on LOAD CAM FILE, then run the loopback test
The control shows the wrong tool	the operation's tool number must be the control's tool memory slot
Parts of the window are cut off	make the window wider, or use it full screen; below 1000 px the operations list moves to its own page